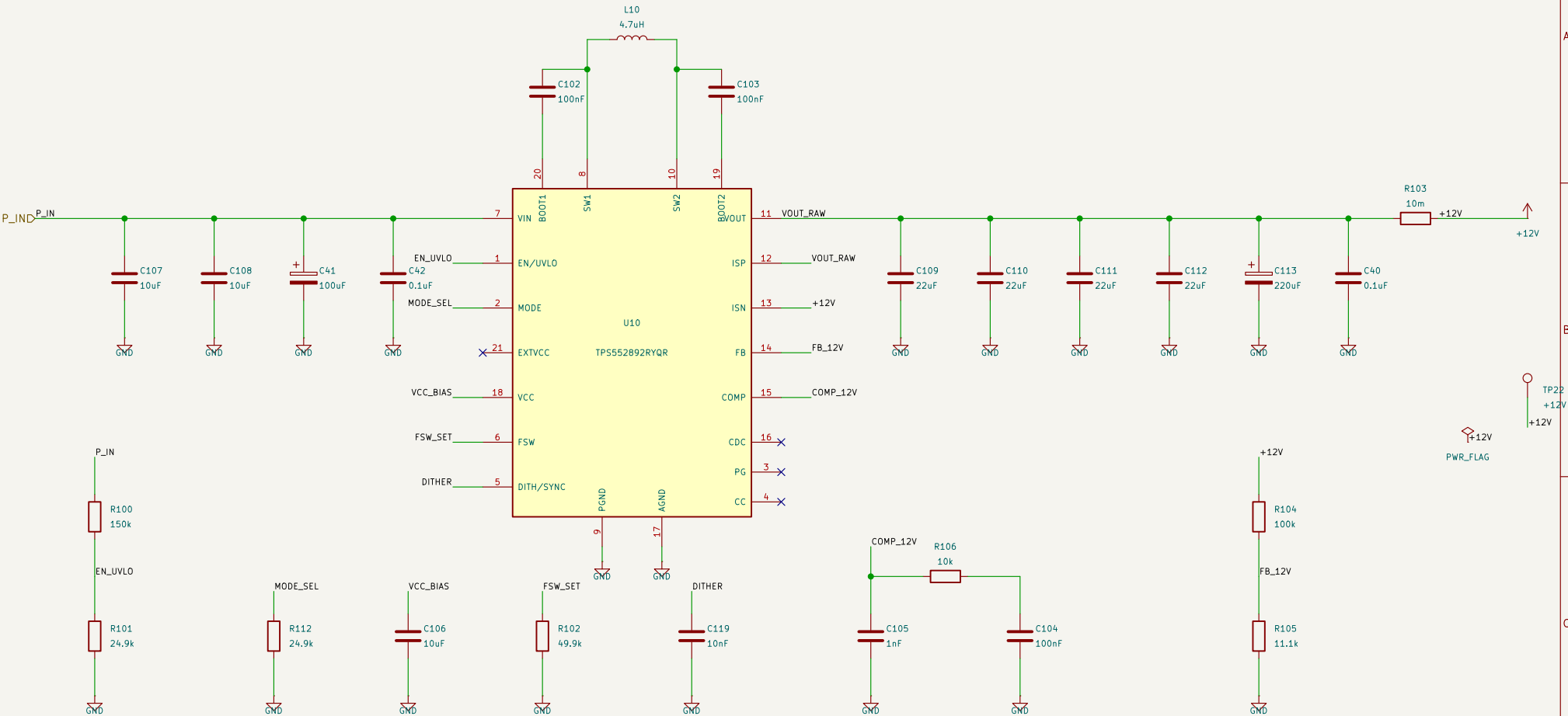


12 V BUCK-BOOST | TPS552892 | 10-14 V INPUT



12.01 V nominal; 3 A continuous design target. 4 A peaks require thermal / transient qualification.

R103: 5 A nominal output limit (4.75-5.25 A tolerance); not a continuous-output rating.

C204 / C214 / C224: 100 uF each. Total connected 12 V capacitance: 709 uF nominal, <851 uF at +20%.

Startup: keep switched loads and downstream rails OFF until 12 V settles; stagger pump / actuator starts.

MODE pulled low; PFM. EXTVCC open; internal bias. PG / CC unused. UVLO retained: -8.64 V on / -7.89 V off.

Layout: tight input/output ceramic loops; Kelvin R103 sense; ground thermal vias. Validate temperature and load steps.

Sheet: /Power/12V/		
File: 12v.kicad_sch		
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KiCad E.D.A. 10.0.4		Id: 8/24